

1 Literature Review

Chronic Kidney Disease (CKD) is a long-term medical condition in which the kidneys gradually lose their ability to filter the blood out of waste. It can be very dangerous to a patient if detected late. There are 5 stages of CKD. Detection of CKD at an early stage is very difficult as the patient shows very few symptoms. Usually, it is detected at advanced stages only, where the ultimate solution is dialysis or a kidney transplant. The most common tests to determine whether a person has CKD or not are eGFR (Estimated Glomerular Filtration Rate), Urine Tests, Blood Tests, and Blood Pressure Tests.

Over the last decade, systems using Artificial Intelligence (AI) and Machine Learning (ML) have gained immense efficiency in the detection of CKD. Nowadays, advanced machine learning models and methods such as Random Forest, XGBoost, ANN, and XAI are being used to predict diseases, helping doctors more effectively.

1.1 Approaches for Feature Selection and Dimensionality

The main and most common dataset repository used is the UCI Machine Learning CKD dataset, having a total of 24 input features. But overall, managing 24 features in the real world is a bit hard as it can have bottlenecks, high costs, and data errors. To overcome the problem of having too many features, researchers have made simple and clean paths to reduce any unwanted extra information.

1. **Supervised Feature Selection:** Al-Jamimi (2024) designed a system using Recursive Feature Elimination connected to an SVM that tests all features and eliminates those which have weak predictive power. Chittora et al. (2021) concluded that basic models may work efficiently but they fail considerably when the data has not been processed. Decision Trees and Naïve Bayes were used, but the handling of missing values was very short, thus decreasing the accuracy of the model. Similarly, Khan et al. (2020) analyzed baseline models and concluded that data preprocessing is one of the major factors which influences the efficiency and accuracy of the models. Shukla and Pillai (2023) and Halder et al. (2024) studied large medical records to find the smallest subset of clinical predictors needed to precisely classify the levels.
2. **Principal Component Analysis (PCA):** It is a method to reduce the number of features of large and complex datasets while maintaining important features. Antony et al. (2021) used unsupervised learning with integrated Pearson correlation, which eliminates multicollinearity features while maintaining crucial medical information.
3. **Extreme Feature Stripping:** As we do not always need large data for accurate predictions, selecting a few vital features can sometimes lead

to good, accurate predictions. Moreno-Sánchez (2023) implemented the XGBoost algorithm and verified that by selecting only three features (Hemoglobin, Specific Gravity, and Hypertension), this model performed just as good as other models which have more features. Islam et al. (2023) also took a quite different but somewhat related approach by taking 30% of the features.

1.2 Development of Machine Learning Model Algorithms

1. **Baseline Models:** These are the foundational models for ML which include KNN, Naïve Bayes, Linear Regression, SVM, and Decision Trees. Chittora et al. (2021) and Khan et al. (2020) analyzed these models and found key insights indicating that the dataset is imbalanced, contains noise and errors, has missing entries, and the models were over-fitted.
2. **Ensemble Methods:** These are modern methods in which more than one method is combined together to get an accurate prediction, including XGBoost, Random Forest, and AdaBoost. Islam et al. (2023) analyzed 12 various models and found out XGBoost had the best accuracy of 98.3%. Halder et al. (2024) tested Random Forest and AdaBoost combined together, which created a stable and reliable model.

1.3 Methods for Improving Model Performance

Metaheuristic Method: This method is inspired by nature to tune the model and enhance its accuracy and performance. Grey Wolf Optimizer (GWO) is a method inspired by how wolves hunt in a pack. They consistently reposition themselves until they catch their prey; similarly, this algorithm adjusts the model until it finds the best answers or classifications.

Ghosh et al. (2025) used this method to fine-tune the eGFR equation which uses creatinine and cystatin C (important indicators). After applying it, the errors in RMSE were reduced by 37.3%. Lei et al. (2022) reviewed decades of ML research and found out that models perform well when external factors or features are controlled wisely.

1.4 Decision Making with Explainable AI (XAI)

These models are accurate, but doctors cannot completely rely on them as they cannot find out how these models reached a particular decision. XAI is a methodology that explains and interprets the decisions made by models to humans in an effective manner, which is more trustworthy.

One major problem with models like ensemble learning is that when an input feature goes in and an output comes out, no one knows how it happened—just like a “black box”. For this, there are solutions like:

1. **SHAP:** A framework used to interpret and explain the decisions made by ML models. Jawad et al. (2025) used Post-Hoc SHAP on ensemble models which explains the decision once it is made by the model, finding that Hemoglobin had a 40% contribution and creatinine had a 30% contribution. Moreno-Sánchez (2023) also used Post-Hoc SHAP and found out that Hemoglobin was one of the major and strongest indicators for the early detection of CKD.
2. **DBRB:** It is an intrinsic model that interprets and explains during the ongoing decision-making process; it is built-in. Zhao et al. (2025) used this, and the diagnosis showed proper step-by-step reasoning.

1.5 Handling Imbalanced and Incomplete Data

In the medical field, data anomalies, inconsistency, missing values, and imbalances are among the most common and major problems for prediction models. Old models majorly used accuracy to determine performance, but it is quite sensitive to noise, errors, and missing values. Jawad et al. (2025) and Khan et al. (2020) used F1-score, Sensitivity, Recall, and AUC-ROC instead of just accuracy, confirming that these classifiers work sufficiently better on imbalanced data.

Modern systems use “imputations”, which is an efficient and smart filling technique to handle the missing data problem. Ekanayake and Herath (2020) verified that imputations help maintain a stable and reliable performance of the model even if there are missing values in the dataset. Meeusen et al. (2022) found that there was a racial bias in the eGFR equation, and they modified the equation to make it fair for all races.

1.6 Specific Research Gaps

- **Feature Selection Problem:** Researchers tried to use as few features as possible to predict CKD faster and more cost-efficiently. But there is one major drawback: models trained over these minimal features might fail in real-world situations. Moreno-Sánchez (2023) designed a model using 3 features, and the results were quite accurate. Shukla and Pillai (2023) also used statistically important features, and Al-Jamimi (2024) used similar related methods, but the limitations were that these models work well only on specific and clean datasets. They do not perform well on diverse classifications of patients. Antony et al. (2021) used PCA, which is a good feature selection methodology, but it has the disadvantage of lacking a proper balance regarding which features to retain.
- **SHAP Problem:** Models can be accurate, but doctors cannot completely rely on them if their inner workings are not known. Techniques like SHAP overcome this problem, but SHAP explains and interprets the model only after a decision is

made, providing an approximation that might not be 100% accurate and trustworthy. Jawad et al. (2025) and Moreno-Sánchez (2023) used SHAP but faced this limitation.

- **Static Model Problem:** CKD develops slowly over months or years. Researchers took only one instance of a dataset or one particular lab result to classify either the stage of CKD or simply binary CKD/Non-CKD status. Santhiya et al. (2024), Islam et al. (2023), Chittora et al. (2021), and Ghosh et al. (2025) used this process, but it has a major limitation: a single lab result cannot detect how fast the disease is spreading over time.
- **Data Problem:** Medical datasets are large, complex, and often incomplete due to missing entries, which degrades model performance. Ekanayake & Herath (2020) and Delrue et al. (2024) verified this problem. Similarly, older eGFR equations contained racial bias, inclining the models inaccurately toward specific demographic outcomes. Meeusen et al. (2022) and Farrell & Vassalotti (2024) highlighted this critical issue.

Table 1: Summary of Existing Literature on Machine Learning Frameworks for Chronic Kidney Disease Prediction

#	Paper Name	Author(s) & Year	Citation (IEEE)	Dataset Used & Footprint	Data Pre-processing	Best Model & Metric	Identified Gaps
1	Machine Learning Framework for Early Detection of CKD Stages Using Optimized eGFR	Ghosh, Widadalla, & Khandoker (2025)	[1] S. K. Ghosh et al., <i>IEEE Access</i> , vol. 13, 2025.	Multi-biomarker cohort (SCr and SCysC).	Metaheuristic population parameter adjustments.	XGBoost + GWO (Reduced Staging RMSE by 37.3%).	Heavily relies on complete laboratory panels; fails with missing markers.
2	A CKD Diagnostic Model Based on An Interpretable Deep Belief Rule Base	Zhao, Zhang, Duan, Che, & Ma (2025)	[2] Y. Zhao et al., <i>IEEE Access</i> , 2025.	Expert rule matrices mapped against clinical lines.	Evidential reasoning rule-base initialization.	Interpretable DBRB (High transparency baseline).	Static expert templates limit adaptive learning of raw variations.
3	A Study on the Application of XAI on Ensemble Models for Predictive Analysis of CKD	Jawad, Verma, Amsaad, & Ashraf (2025)	[3] K. M. T. Jawad et al., <i>IEEE Access</i> , vol. 13, 2025.	Tabular clinical indicators; binary target.	Missing value handling and standard feature scalers.	Optimized Ensemble + SHAP (Peak transparency).	Post-hoc explanations (SHAP) are mere approximations of the core.
4	ML-CKDP: Machine learning-based CKD prediction with smart web application	Halder et al. (2024)	[4] R. K. Halder et al., <i>J. Pathol. Inform.</i> , vol. 15, 2024.	ML-CKDP clinical records dataset footprint.	Multi-stage cleaning and outlier threshold clipping.	Hybrid RF-AdaB Ensemble (Maximized stability).	Vulnerable to extreme variance during volatile laboratory spikes.
5	Synergistic Feature Engineering and Ensemble Learning for Early Chronic Disease Prediction	Al-Jamimi (2024)	[5] H. A. Al-Jamimi, <i>IEEE Access</i> , vol. 12, 2024.	Benchmark medical database arrays.	Recursive Feature Elimination wrapper engine.	SVM-RFE + XGBoost (Reduced computational footprint).	Label-dependent feature subsets fail on highly diverse classifications.
6	Screening, identifying, and treating CKD: why, who, when, how, and what?	Farrell & Vassalotti (2024)	[6] D. R. Farrell and J. A. Vassalotti, <i>BMC Nephrol.</i> , vol. 25, 2024.	Clinical epidemiological patient staging guidelines.	C-G-A risk stratification charting guidelines.	C-G-A Stratification Protocol (Gold standard baseline).	Lacks an automated machine learning script; purely clinical.
7	Data-Driven Early Diagnosis of CKD: Development and Evaluation of an Explainable AI Model	Moreno-Sánchez (2023)	[7] P. A. Moreno-Sánchez, <i>IEEE Access</i> , vol. 11, 2023.	Extreme attribute reduction framework.	Aggressive target-driven variable trimming.	3-Feature XGBoost (Hemoglobin, SG, Hypertension).	Drastic feature compression loses vital medical variance.
8	CKD Prediction Using ML Algorithms and the Important Attributes for the Detection	Shukla & Pillai (2023)	[8] G. Shukla and S. K. Pillai, <i>IEEE GlobConET</i> , 2023.	Standard multi-attribute benchmark dataset.	Feature importance ranking filters.	Optimized Feature Classifier (Minimized overhead).	Over-fits to static snapshot data; lacks deployment evaluation.
9	Comprehensive Performance Assessment of Deep Learning Models in Early Prediction of CKD	Akter et al. (2021)	[9] S. Akter et al., <i>IEEE Access</i> , 2021.	Longitudinal, sequential patient records.	Sequential multi-attribute time-field restructuring.	Bidirectional LSTM & GRU (Superior temporal processing).	Fails to integrate aggressive feature optimization with sequential deep models.
10	A Comprehensive Unsupervised CKD Prediction Framework	Antony et al. (2021)	[10] L. Antony et al., <i>IEEE Access</i> , 2021.	Unlabeled patient health profiles.	PCA & Pearson Correlation.	PCA-Driven Unsupervised Pattern Classifier .	PCA transformation hides which specific clinical features to retain.
11	Prediction of Chronic Kidney Disease - A Machine Learning Perspective	Chittora et al. (2021)	[11] P. Chittora et al., <i>IEEE Access</i> , vol. 9, 2021.	Tabular medical health parameter repositories.	Missing value imputation, noise handling, data balancing.	Optimized Baseline Classifiers (Evaluating imbalances).	Highly sensitive to noise; struggles with high structural missingness.
12	An Empirical Evaluation of ML Techniques for Chronic Kidney Disease Prophecy	Khan et al. (2020)	[12] B. Khan et al., <i>IEEE Access</i> , vol. 8, 2020.	Standard repository datasets with missing points.	Basic column cleaning and dataset normalization filters.	Shallow Baseline Trees (With robust metrics).	Models remain static; high error rates under heavy data imbalances.
13	Chronic Kidney Disease Prediction Using Machine Learning Methods	Ekanayake & Herath (2020)	[13] I. U. Ekanayake and D. Herath, <i>Proc. IEEE MER-Con</i> , 2020.	Heterogeneous and incomplete medical records.	Advanced statistical missing data imputations.	Imputation-Stabilized Tree Models .	Overlooking class-balancing causes dips on diverse datasets.
14	Chronic Kidney Disease Prediction Using Machine Learning Algorithms	Santhiya et al. (2024)	[14] K. Santhiya et al., <i>IEEE IC-IETDW</i> , 2024.	Static laboratory check-up entries.	Standard attribute filtering and label encoding.	Shallow Ensemble Baseline (Optimized checkpoint vectors).	Utilizes a single point-in-time value, missing long-term progression.
15	Chronic kidney disease prediction based on machine learning algorithms	Islam et al. (2023)	[15] M. A. Islam et al., <i>J. Pathol. Inform.</i> , vol. 14, 2023.	Compressed dataset (30% feature selection footprint).	Matrix compression and mean/mode imputation.	XGBoost Classifier (Achieved peak accuracy of 98.3%).	Relies entirely on static configurations; fails to account for temporal progression.